ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: Hema Varshini K

Roll No: 26CS1146

Date:

Subject Name: Professional English – I (Lab Integrated)

SECTION 2: BIOGRAPHICAL NOTE – SATYA NADELLA

- **Full Name:** Satya Narayana Nadella
- **Date of Birth:** 19 August 1967
- **Birthplace:** Hyderabad, India
- **Nationality:** American
- **Profession:** Engineer and business executive
- **Company:** Microsoft
- **Position:** Chairman and Chief Executive Officer (CEO)

Early Life and Education

Satya Nadella was born in Hyderabad, India. He studied at Hyderabad Public School and completed his bachelor's degree in electrical engineering at Manipal Institute of Technology. Later, he moved to the United States for higher education. He earned a master's degree in computer science from the University of Wisconsin–Milwaukee and an MBA from the University of Chicago.

Career and Achievements

- **Joined Microsoft:** He joined Microsoft in 1992.
- **Became CEO:** He was appointed CEO of Microsoft in February 2014.
- **Cloud Computing:** He played an important role in expanding Microsoft's cloud computing business, particularly Microsoft Azure.
- **Artificial Intelligence:** He has helped Microsoft expand its work in AI technologies and products.
- **Leadership:** He encourages teamwork, innovation, continuous learning, and collaboration.
- **Author:** He wrote *Hit Refresh*, a book about his personal journey, leadership, and Microsoft's transformation.

Personal Interests and Qualities

- He has a strong interest in technology and innovation.
- He enjoys reading books and learning new things.
- He has spoken about his interest in cricket and how team sports teach leadership.
- He values empathy, respect, and understanding other people's needs.
- He believes people should develop a growth mindset and keep learning throughout their lives.

Conclusion & Inspiration

Satya Nadella inspires students because he demonstrates the importance of education, hard work, patience, and continuous improvement. His journey from Hyderabad to becoming the leader of a major global technology company shows how technical knowledge and leadership skills can create opportunities. He is a successful technology leader whose contributions have influenced the development of cloud computing and artificial intelligence. His leadership teaches us to be curious, work as a team, understand others, and never stop learning.

Reference: Microsoft official biography — https://news.microsoft.com/source/exec/satya-nadella/

Hemavarshini K

Email: nalinisuriya2006@gmail.com | LinkedIn: linkedin.com/in/hemavarshini-k-858b8b43b
Student at St. Joseph's College Of Engineering, Chennai, Tamil Nadu, India

Professional Summary

First-year B.E. Computer Science and Engineering student with strong foundational skills in C programming and analytical problem-solving. Deeply interested in Cloud Infrastructure, operating systems, and networking. Aspiring to build a career as an Infrastructure or Cloud Engineer focused on designing reliable, scalable, and highly available background systems.

Education

St. Joseph's College Of Engineering — *BE CSE* (August 2026 - May 2030)

National Matriculation Higher Secondary School — *Bio-Maths* (2025 - 2026)

Core Focus & Skills

- **Programming:** Foundational C Programming, algorithmic problem solving
- **Systems & Infrastructure:** Operating Systems concepts, Networking basics
- **Interests:** Cloud Architecture, Distributed Systems, Scalable Infrastructure

Recommended Cloud Engineering Learning Roadmap

As a first-year engineering student, building a progressive roadmap ensures deep mastery over 4 years:

Year 1: Foundations & Systems
Focus heavily on college fundamentals: Advanced data structures in C/C++, GNU/Linux command-line basics, and core Computer Networking concepts (TCP/IP, OSI model, DNS).

Year 2: Scripting & Cloud Introduction
Pick up an industry-standard scripting language like Python. Learn the absolute basics of public cloud platforms by aiming for an entry-level certification.

Year 3: DevOps & Infrastructure as Code (IaC)
Dive into Git version control, containerization using Docker, and automated infrastructure deployment using Terraform.

Year 4: Advanced Architecture & Capstone
Design and deploy scalable web apps, practice serverless computing, and complete a high-impact cloud capstone project.

Target Entry-Level Cloud Certifications

- **AWS Certified Cloud Practitioner (CLF-C02)** — Ideal starting point to learn core AWS global infrastructure services.

PROFESSIONAL ENGLISH I | BATCH B

My SWOC: Claude and Cloud Infrastructure

Target role: Infrastructure / Cloud Engineer

Goal: build reliable, scalable cloud systems that power AI tools like Claude

Hemavarshini K | 1st Year B.E. CSE | EN25C09

SWOC Matrix

S | Strengths

C, problem solving, quick learning

W | Weaknesses

Only one language (C)

O | Opportunities

CodeChef / HackerRank contests; growing demand for tech skills

C | Challenges

Fast-changing tech; strong competition

Hemavarshini K | EN25C09

Strengths

C programming

Used to build operating systems, the base layer of all infrastructure.

Problem solving

Helps me find solutions to technical problems step by step.

Quick learning

Helps me adapt to new technologies and tools.

Logical thinking

Helps me understand problems clearly and develop systematic solutions.

Weaknesses

Learning pace

I sometimes need more time to understand complex technical concepts.

Experience gap

I know only one language, C (short term). I plan to learn Python and other languages.

Opportunities

Contests

CodeChef and HackerRank contests.

Growing demand

AI tools like Claude run on cloud infrastructure and need cloud engineers.

NPTEL courses

Programming, networking and cloud concepts.

Hackathons and clubs

Practical experience and projects.

Challenges

Fast-changing technology

Tools and skills keep evolving.

Strong competition

Many students target the same roles.

Workload balance

Managing academics and skill development.

Limited experience

Few practical projects as a first-year student.

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Action Plan: Using Opportunities

Monthly contest

One CodeChef contest every month.

NPTEL

Courses in programming and networking.

Hackathons and clubs

Hands-on teamwork and projects.

Claude API project

Planned.

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4-Year Milestones

Year	Primary focus	Target deliverable	Target credential
Year 1	C, OS, Python	Python mini project	NPTEL certificate
Year 2	Python, Linux	Cloud basics	Cloud fundamentals certificate
Year 3	DevOps, databases	Internship	Cloud certification
Year 4	Advanced cloud and infrastructure	Major project + placement	Placement prep, cloud certification

What I Learned

Write & Improve – University of Cambridge

Write & Improve is a free online English writing practice tool developed by the University of Cambridge. It helps learners improve their English writing skills through writing activities and automatic feedback.

Key Things I Learned

1. **Improve Writing Skills:** I learned that I can improve my English writing by practising different writing tasks.
2. **Different Writing Activities:** The website provides hundreds of writing tasks for different English proficiency levels.
3. **Automatic Feedback:** After submitting my writing, the website automatically checks it and gives feedback on areas that need improvement.
4. **CEFR Levels:** I learned that my writing performance can be assessed according to CEFR (Common European Framework of Reference for Languages).
5. **Correct and Rewrite:** I can read the feedback, correct my mistakes, rewrite my sentences and submit my work again.
6. **Track Progress:** By creating an account, I can save my writing and practise more activities.

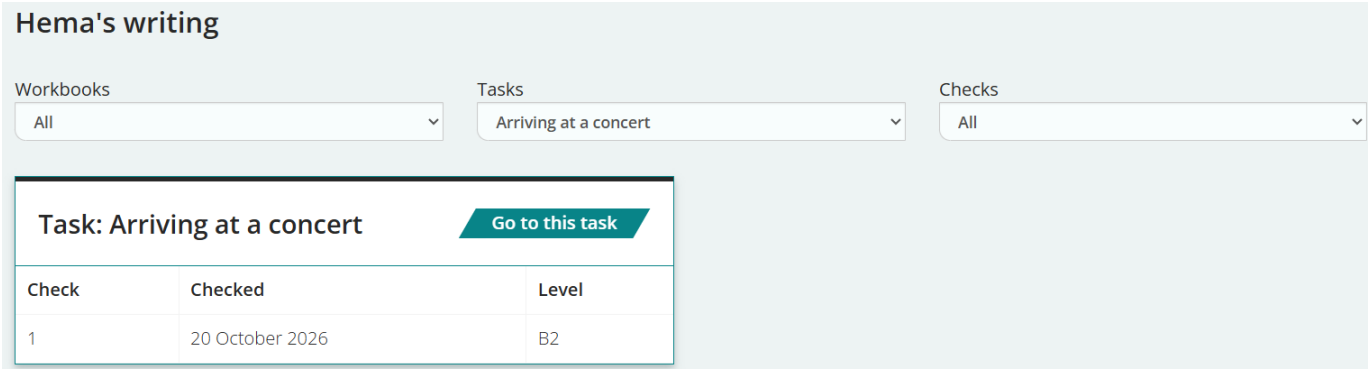
What Skills Did I Develop?

- Improved English grammar and sentence formation.
- Learned to organise ideas into paragraphs.
- Improved vocabulary and spelling.
- Understood how to write more clearly.
- Learned to identify and correct writing mistakes.
- Developed confidence in English writing.

My Overall Learning

I learned that Write & Improve is a useful platform for developing English writing skills. It helps me identify my mistakes, understand where I need improvement and practise writing regularly. I can use this website to become a better and more confident English writer.

Conclusion: This website taught me that regular practice, feedback and correction are important for improving my English writing skills.


Hema's writing
Workbooks
All
Tasks
Arriving at a concert
Checks
All
Task: Arriving at a concert
Go to this task
Check
Checked
Level
1
20 October 2026
B2

MY LEARNING EXPERIENCE

Speak & Improve – English Speaking Practice

WEBSITE	speakandimprove.com

INTRODUCTION

Speak & Improve is an online English-speaking practice platform developed by the University of Cambridge. It helps learners practise speaking through interactive exercises and receive automated feedback. I used this platform to practise English and understand my strengths and areas for improvement.

KEY LEARNINGS

01 Speaking Confidence

I learned to express my thoughts in English with greater confidence and to overcome my fear of making mistakes.

02 Pronunciation and Clarity

I understood the importance of clear pronunciation and speaking in a way that others can easily understand.

03 Fluency and Communication

I learned that regular speaking practice helps me communicate more naturally, smoothly, and effectively.

04 Vocabulary Development

I understood how learning new words and using them in sentences helps me express my ideas more clearly.

05 Self-Assessment

I learned how to evaluate my speaking performance, identify my weaknesses, and focus on areas that need improvement.

SKILLS DEVELOPED

Skill	Improvement
Communication	Expressing ideas clearly
Confidence	Speaking without hesitation
Pronunciation	Speaking words clearly
Fluency	Speaking more naturally
Self-learning	Identifying areas for improvement

PERSONAL REFLECTION

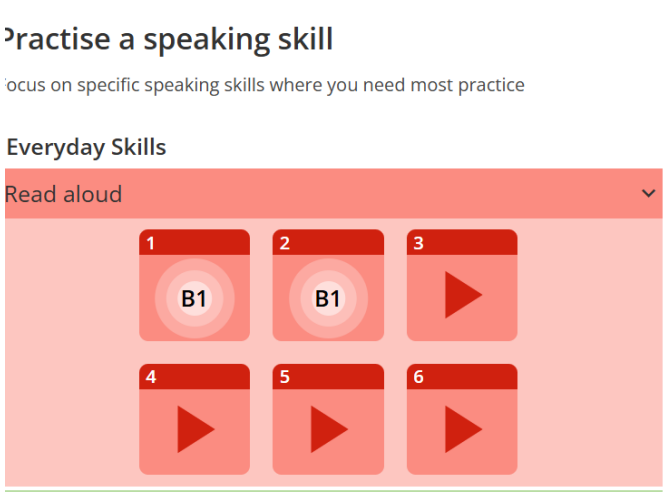
This learning activity helped me understand that effective communication is important for academic and professional success. I realized that making mistakes is a natural part of learning and that consistent practice is the key to improvement.

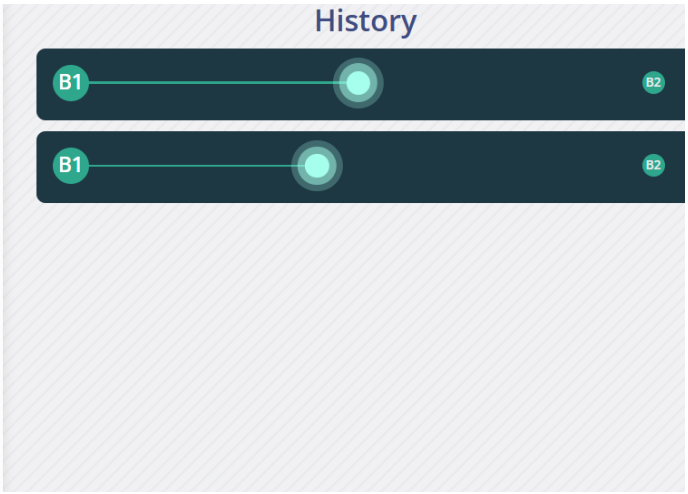
CONCLUSION

Speak & Improve gave me an opportunity to practise English independently and assess my speaking skills. I will continue using this platform to improve my fluency, pronunciation, and confidence. My goal is to communicate effectively in interviews, presentations, and everyday situations.

MY LEARNING GOAL

Practise English speaking every day and gradually improve my communication skills.





RESQMESH

Enhanced Project Summary & Strategic Blueprint

1. The Core Challenge & Value Proposition

Infrastructure Vulnerability: Disasters disrupt cellular networks and internet backhaul, creating a critical bottleneck precisely when people need to exchange life-saving status updates.

Graceful Degradation: Instead of losing all connectivity, ResQMesh converts existing smartphones into a resilient, decentralized fallback layer to keep local communication alive.

Zero Infrastructure Dependency: It requires no specialized emergency hardware, cell towers, or public internet access to exchange critical data locally.

2. Technical Architecture & Core Mechanics

- **Dynamic Peer Discovery:** Nearby devices automatically establish temporary local links using Wi-Fi Direct/P2P and Bluetooth Low Energy (LE).
- **Intelligent Store-and-Forward Routing:** Messages are saved securely in a local database and passed hop-by-hop as users move through the affected area.

3. Business Model & Operational Sustainability

ResQMesh utilizes a freemium framework that aligns widespread public safety features with a recurring institutional enterprise engine:

Democratic Safety Core (Free Tier)

The foundational emergency capabilities—including offline messaging, peer discovery, and local alerts—remain entirely free for public use.

Institutional Monetization (B2B/B2G Paid Tier)

Revenue is generated via recurring subscriptions from campuses, corporations, and volunteer groups for premium institutional management tools:

- Centralized administrative incident dashboards and spatial analytics.
- Managed communication groups and targeted emergency admin broadcasts.
- Deployment services, customized workflows, security audits, and live emergency drills.

4. Target Metrics & Execution Roadmap

Key Success Indicators

- **Message Reach:** Percentage of test messages successfully delivered within a defined time window.

- **Delivery Latency:** Median time elapsed from sender dispatch to recipient acknowledgement.
- **Network Resilience:** Real-time network throughput and packet survival as standard nodes are dynamically removed.
- **Battery Efficiency:** Minimizing the energy footprint consumed per hour of active mesh participation.

Phased Rollout Schedule

- **Phase 1 (Month 1):** Clickable UI, local database schema definition, and baseline peer discovery scripts.
- **Phase 2 (Month 2):** Wi-Fi P2P prototype integration featuring basic store-and-forward routing.
- **Phase 3 (Months 2–4):** Controlled 20–50 hardware device live pilot focusing explicitly on battery drain profiles and link reliability.
- **Phases 4–5 (Months 4–12):** Hardened cryptographic layer, administrator enterprise dashboard launch, and cross-platform production expansion.

Immediate Near-Term Milestone: Successfully deliver short text messages across a local mesh of 5 to 10 independent mobile devices after intentionally disabling all outbound internet and cellular access.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

Enhancing Soft Skills and Personality

NPTEL Course Summary (Weeks 1–8)

Instructor: Prof. T. Ravichandran | IIT Kanpur

Week 1: Mindset and Core Frameworks

- **Course Foundation:** Introduces the core concepts of soft skills, highlighting how behavioral change, preparedness, and receptivity drive self-improvement.
- **Mindset Types:** Explains the distinct differences between a Fixed Mindset and a Growth Mindset.
- **Growth Progression:** Shares actionable strategies to move past psychological barriers and foster a continuous learning mindset.

Week 2: Time Management and Assertiveness

- **Perception of Time:** Analyzes how individuals perceive time and the importance of personal efficiency.
- **Beating Procrastination:** Identifies the root causes of delay and offers practical tactics to overcome laziness.
- **Assertive Communication:** Focuses on establishing boundaries, explicitly teaching students how to stop saying "yes" just to appease others.

Week 3: Emotional Regulation and Positive Thinking

- **Social Navigation:** Covers different types of human personalities and how to say "no" constructively.
- **Anger Management:** Provides emotional intelligence strategies to intercept, control, and channel anger into productivity.
- **Optimism:** Focuses on gaining personal power and resilience through deliberate positive thinking patterns.

Week 4: Social Dynamics and Likability

- **Interpersonal Chemistry:** Examines behavioral traits and habits that unintentionally alienate or cause others to dislike you.
- **Building Rapport:** Discusses the psychology of what makes a person genuinely likable and approachable.
- **Personal Attraction:** Explores how to project an attractive, charismatic personality through both verbal and non-verbal cues.

Week 5: Overcoming Common Behavioral Errors

- **Pitfall Identification:** A deep dive into standard behavioral errors, communication missteps, and flawed habits that hinder professional growth.
- **Habit Reconstruction:** Outlines how to identify systemic individual flaws and replace unhealthy routines with constructive ones.

Week 6: Humour and Financial Management

- **Workplace Humor:** Covers the function of healthy humor in professional communication to reduce stress and bridge connection gaps.
- **Money and Identity:** Explores how personal psychology shapes your spending and saving habits.
- **Financial Literacy:** Delivers foundational advice on managing money responsibly as an element of stability.

Week 7: Holistic Health and Relationships

- **Physical Well-being:** Connects physical fitness directly to personality stamina, emphasizing regular exercise, sleep hygiene, and diet.
- **Emotional Well-being:** Discusses the psychological dimensions of love, emotional bonds, and maintaining healthy interpersonal relationships.

Week 8: Professional Etiquette and Cognitive Skills

- **Business Conduct:** Details workplace ethics, professional etiquette, and corporate social graces.
- **Cognitive Enhancements:** Provides memory retention techniques and mental exercises to optimize memory performance.
- **Environmental Mindfulness:** Concludes with an emphasis on environmental care as part of a well-rounded, ethical character.

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Hemavarshini K

Email: hema.120109@gmail.com | GitHub: github.com | CodeChef: codechef.com

About Me

I'm a student passionate about C programming, data structures, and problem-solving. I enjoy exploring how programs work internally and building efficient solutions for real-world challenges.

- **Focus:** C programming and core DSA concepts
- **Currently:** Building projects and solving coding problems
- **Goal:** Strengthen my programming fundamentals and algorithmic thinking

What I'm Learning

Languages & Tools: C, Git, GitHub, VS Code, CodeChef

Topics: Arrays & Strings, Linked Lists, Stacks & Queues, Trees & Graphs, Sorting & Searching, Recursion & Backtracking, Memory Management, File Handling

Projects

Student Record Management System

A console-based application designed to manage student databases, focusing on structured data layout and persistent storage.

- **Skills used:** C, File I/O, Structs, Pointers, Memory Management

Custom Stack & Queue Implementation

A core data structures library built from scratch to handle dynamic memory allocation, demonstrating deep algorithmic logic.

- **Skills used:** C, Linked Lists, Dynamic Memory (malloc/free), Pointers

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Hemavarshini k

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Chennai, Tamil Nadu, India

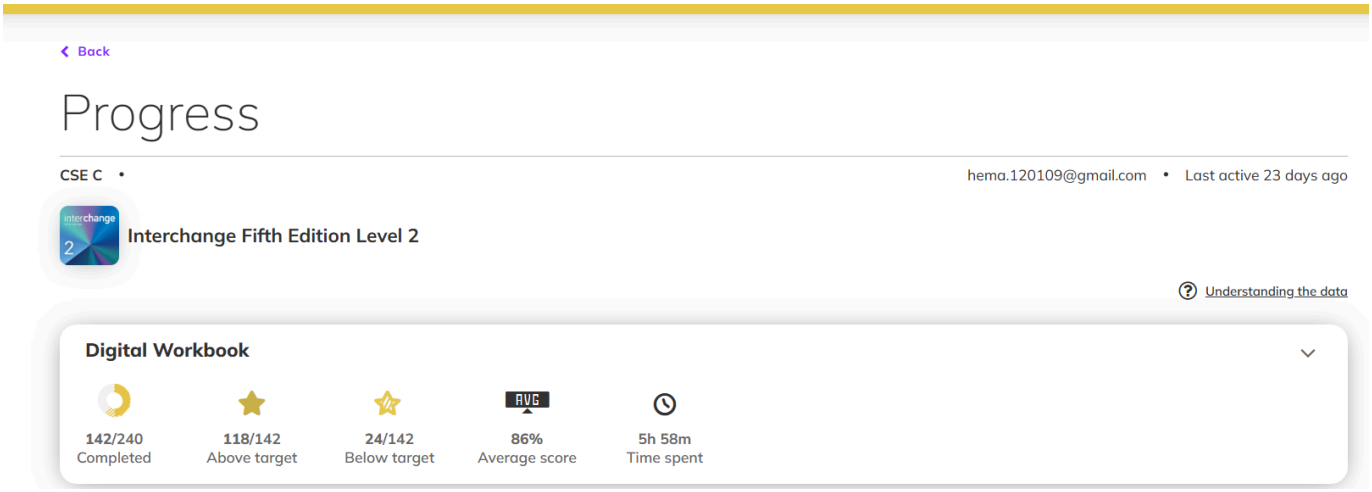
Summary

Hello, I am Hemavarshini K, a first-year B.E. Computer Science and Engineering student. My key strengths are C programming and problem solving. I am interested in Cloud infrastructure, the systems that run quietly behind every app and service. I am building a strong base in programming, operating systems and networking. My goal is to become an infrastructure or cloud engineer who designs reliable, scalable systems. I learn quickly, and I am ready to take on new challenges.

Education

St. Joseph's College Of Engineering
BE CSE · (August 2026 - May 2030)

National matriculation higher secondary school
Bio-Maths · (2025 - 2026)


Back
Progress
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hema.120109@gmail.com • Last active 23 days ago
Interchange Fifth Edition Level 2
Understanding the data
Digital Workbook
142/240
Completed
118/142
Above target
24/142
Below target
AVG
86%
Average score
5h 58m
Time spent

Why the Sky is Blue

The Story

Mira was lying awake in her bed, struggling to fall asleep. Feeling curious, she turned to her grandmother and asked, "Grandma, why is the sky actually blue?"

Grandma smiled warmly and explained, "Although sunlight looks clear to us, it is actually a combination of all the colors of the rainbow hidden inside it—including red, yellow, green, and blue."

"When this brilliant light reaches Earth, it collides with the gases and particles in our atmosphere. Because blue light travels in shorter waves, it scatters in every direction much more than the other colors."

"So that is why we see a blue sky!" Mira exclaimed.

"Exactly," said Grandma. "That is the scientific reason behind it."

Mira let out a gentle yawn, feeling satisfied with the answer. "Goodnight, blue sky," she whispered.

"Goodnight, little scientist," replied Grandma softly.

CEFR Level Assessment

Overall Level: B1 (Intermediate)

The text is accessible and clear, making it a great reading passage for B1 language learners.

Breakdown by Linguistic Features

Vocabulary	Uses descriptive and slightly advanced words like struggling, curious, combination, brilliant, collides, particles.	B1–B2
Grammar & Sentence Structure	Employs past tense narratives and conversational structures suitable for intermediate comprehension.	B1

St. Joseph's College of Engineering

OMR, Chennai – 119 (An Autonomous Institution)

Professional English – I (Lab Integrated)

Model – I – Practice Worksheet: Questions with Answers

Course Code: EN25CO9

PART A

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets.

1. Neither the project supervisor nor the lead programmers _____ (recognize / recognizes) the anomaly in the build script.
   Ans: recognize – with neither…nor, the verb agrees with the nearer subject ("programmers", plural).
2. Each of the automated sorting arms _____ (require / requires) recalibration after twelve hours of continuous duty.
   Ans: requires – "Each" is singular.
3. Either the primary sensor or the auxiliary telemetry units _____ (transmit / transmits) packet data to the ground station.
   Ans: transmit – the nearer subject "units" is plural.
4. The consortium of aerospace contractors _____ (agree / agrees) to adhere to the revised tolerance levels.
   Ans: agrees – "consortium" is a collective noun acting as one unit.
5. Physics and material science _____ (form / forms) the core theoretical foundation for nanoscale fabrication.
   Ans: form – two subjects joined by "and" take a plural verb.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs.

1. Last month, the metallurgy division _____ (conduct) destructive fatigue testing on the titanium alloy shaft.
   Ans: conducted
2. The field engineers _____ (deploy) thirty IoT water-monitoring nodes across the river basin.
   Ans: deployed
3. During the pressure overload trial, the relief valve _____ (open) at 150 psi as designed.
   Ans: opened
4. The research unit _____ (publish) their initial benchmark results on low-power RISC-V architectures.
   Ans: published
5. In the 2025 dry run, the autonomous navigation algorithm _____ (detect) all dynamic road obstacles successfully.
   Ans: detected

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets.

1. Gallium nitride (GaN) semiconductors operate at _____ (high) power efficiency than standard silicon transistors.
   Ans: higher
2. Solid-state batteries deliver a _____ (safe) operational threshold under rapid discharge cycles than traditional lithium cells.
   Ans: safer
3. The revised compiler generates code that executes _____ (fast) than the legacy unoptimized routine.
   Ans: faster
4. Brushless DC motors are considerably _____ (durable) than brushed equivalents due to reduced mechanical wear.
   Ans: more durable
5. The liquid cooling loop maintained a _____ (low) core temperature than the dual fan-sink array.
   Ans: lower

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to).

1. Every operator _____ wear anti-static wrist straps before handling open CMOS motherboards.
   Ans: must (also: has to)
2. Hazardous chemical waste _____ be neutralized prior to disposal into the industrial drain network.
   Ans: must (also: has to)
3. Field technicians _____ log their entry time into the high-voltage testing bay without fail.
   Ans: have to (also: must)
4. To maintain ISO certification, calibration logs _____ be audited biannually by an external inspector.
   Ans: must (also: have to)
5. The test rig _____ undergo an emergency stop inspection before initiating the high-speed turbine run.
   Ans: must (also: has to)

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal (can, could, may, might).

1. With the upgraded DSP chip, the system _____ process 4K video streams in real time.
   Ans: can
2. If thermal throttling occurs, the processing clock rate _____ drop by as much as 30%.
   Ans: may / might / could – possibility.
3. Advanced laser profilometry _____ measure surface roughness down to sub-micron tolerances.
   Ans: can
4. Unshielded signal lines _____ pick up electromagnetic interference from nearby induction coils.
   Ans: can / may
5. The modular chassis design _____ accommodate up to four secondary expansion cards.
   Ans: can

Exercise 6: Relative Clauses

Combine each pair of sentences into a single sentence using a suitable relative pronoun (who, which, that, whose).

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.
   Ans: The civil engineer **who** designed the suspension cable bridge received the national infrastructure award.
2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.
   Ans: The spectrometer, **which** analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.
   Ans: The technical specialist **whose** patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.
   Ans: This microcontroller, **which** features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.
   Ans: The field researchers **who** surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets.

1. If the bus voltage _____ (exceed) 24V, the zener diode will divert the surge to ground.
   Ans: exceeds
2. The watchdog timer will reset the CPU if the main thread _____ (fail) to ping it within 500 ms.
   Ans: fails
3. If the operator _____ (engage) the safety latch, the hydraulic press will not actuate.
   Ans: engages
4. If temperature levels _____ (rise) above 90°C, the cooling fans will switch to maximum RPM.
   Ans: rise
5. The system will trigger an auditory alarm if the toxic gas detector _____ (register) more than 20 ppm.
   Ans: registers

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle).

1. If the backup generator _____ (start) immediately, the data centre servers would not have shut down.
   Ans: had started
2. The turbine bearing would not have seized if the maintenance crew _____ (lubricate) it as scheduled.
   Ans: had lubricated
3. If the security team had patched the zero-day exploit earlier, the data breach _____ (prevent).
   Ans: would have been prevented
4. The circuit breaker _____ (trip) safely if the overcurrent sensor had functioned properly.
   Ans: would have tripped

5. If the telemetry link _____ (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.
   Ans: had remained

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb).

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.
   Ans: The mechanics installed dual damper springs **to isolate** chassis resonance.
2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.
   Ans: Technicians purge the fuel line with dry nitrogen gas **to remove** residual flammable vapors.
3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.
   Ans: We introduced an optical encoder **to determine** spindle velocity accurately.
4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.
   Ans: The developers wrote an automated load testing script **to simulate** ten thousand concurrent connections.
5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.
   Ans: Gold plating is applied to terminal connectors **to prevent** atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition.

1. The aerodynamic team improved aircraft fuel economy by _____ (redesign) the winglet profile.
   Ans: redesigning
2. Always disconnect the power supply before _____ (replace) defective surface-mount components.
   Ans: replacing
3. The facility was praised for _____ (reduce) volatile organic chemical emissions by 40%.
   Ans: reducing
4. Avoid _____ (clean) precision optical lenses with abrasive industrial solvents.
   Ans: cleaning (“avoid” takes a gerund directly).
5. The digital signal processor isolates target audio signals by _____ (filter) out background ambient noise.
   Ans: filtering

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice.

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.
   Ans: 50 mL of hydrochloric acid **was poured** into the titration flask.
2. The automation unit tested the robotic arm under a 50 kg dynamic payload.
   Ans: The robotic arm **was tested** under a 50 kg dynamic payload.
3. Field engineers calibrated the optical pyrometer prior to ignition.
   Ans: The optical pyrometer **was calibrated** prior to ignition.
4. The technicians observed severe cavitation damage along the impeller blades.
   Ans: Severe cavitation damage **was observed** along the impeller blades.
5. We store volatile solvents in flame-resistant containment cabinets.
   Ans: Volatile solvents **are stored** in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverb derived from the adjective in brackets.

1. The stepper motor rotates _____ (precise) in 1.8-degree angular increments.
   Ans: precisely
2. The telemetry ground station _____ (periodic) polls all deployed marine buoy sensors.
   Ans: periodically
3. The thermal imaging camera registered the temperature drop _____ (rapid) during quenching.
   Ans: rapidly
4. Structural joints must be welded _____ (firm) to withstand heavy cyclic vibrational stress.
   Ans: firmly
5. The autonomous drone traversed the confined pipeline corridor _____ (smooth).
   Ans: smoothly

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech.

1. The quality manager asked the technician, “Has the insulation resistance test been documented?”
   Ans: The quality manager asked the technician **whether the insulation resistance test had been documented**.
2. The lead architect enquired, “Can the server cluster handle double the transaction throughput?”
   Ans: The lead architect enquired **whether the server cluster could handle double the transaction throughput**.
3. The field auditor asked the safety engineer, “Did you inspect the gas leakage relief lines?”
   Ans: The field auditor asked the safety engineer **whether he/she had inspected the gas leakage relief lines**.
4. The client asked the team, “Will the firmware update fix the intermittent battery drain?”
   Ans: The client asked the team **whether the firmware update would fix the intermittent battery drain**.
5. The instructor asked the lab group, “Why did the multimeter show fluctuating voltage readings?”
   Ans: The instructor asked the lab group **why the multimeter had shown fluctuating voltage readings**.

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly.

1. The procurement officer purchased several new machineries and equipments for the workshop.
   Ans: Incorrect: *machineries, equipments*. Correct: The procurement officer purchased **new machinery and equipment** for the workshop (or “several new pieces of machinery and equipment”).
2. The laboratory generated a huge amount of test datas during the wind tunnel trials.
   Ans: Incorrect: *datas*. Correct: The laboratory generated a huge amount of test **data** during the wind tunnel trials.
3. We need three more softwares licenses to complete the finite element analysis on the model.
   Ans: Incorrect: *softwares*. Correct: We need three more **software** licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable informations regarding the soil composition.
   Ans: Incorrect: *informations*. Correct: The field report provided **a great deal of valuable information** (or “numerous valuable pieces of information”) regarding the soil composition.
5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.
   Ans: Incorrect: *glasswares* (and the uncountable usage of apparatuses). Correct: All chemical **apparatus and glassware** must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form.

1. Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.
   Ans: worse
2. The redesigned microstrip antenna achieved a _____ (good) signal-to-noise ratio than the baseline patch.
   Ans: better
3. The autonomous vehicle traveled 50 km _____ (far) on a single charge after the weight reduction overhaul.
   Ans: farther / further
4. Switching to asynchronous task queues resulted in _____ (little) CPU idle time during peak batch processing.
   Ans: less
5. We require _____ (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.
   Ans: further ("father" is not a comparative).

PART B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following descriptions in the Simple Past Tense.

1. "The technician inspects the solar arrays and records the peak power conversion value."
   Ans: The technician **inspected** the solar arrays and **recorded** the peak power conversion value.
2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."
   Ans: The chemical analyzer **detected** trace mercury levels and **sent** an automated alert to the environmental dashboard.
3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."
   Ans: Maintenance crews **lubricated** the heavy conveyor bearings and **replaced** the worn rubber belts before the morning shift.
4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."
   Ans: The supervisory software **polled** field sensors every five milliseconds and **archived** temperature trends into a secure database.
5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."
   Ans: The testing team **applied** a 500-volt dielectric pulse across the stator winding and **monitored** for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft two technical comparative sentences for each prompt.

1. Cloud Storage vs. On-Premise Physical Servers (initial deployment cost, elastic scalability).
   Ans: Cloud storage has a **lower** initial deployment cost than on-premise physical servers because no hardware must be purchased upfront. It is also **more elastic** than physical servers, since capacity can be scaled up or down within minutes.
2. Fiber-Optic vs. Copper Twisted-Pair Cables (bandwidth, EMI susceptibility).
   Ans: Fiber-optic cables offer a **much higher** data transfer bandwidth than copper twisted-pair cables. They are also **far less susceptible** to electromagnetic interference because they transmit light instead of electrical signals.
3. Electric Vehicles vs. ICE Vehicles (tailpipe emissions, purchase price).
   Ans: Electric vehicles produce **lower** tailpipe emissions than ICE vehicles, since they release none at all. However, EVs generally have a **higher** initial purchase price than comparable ICE vehicles.
4. Hydraulic vs. Electromechanical Actuators (force output, fluid leakage risk).
   Ans: Hydraulic actuators deliver a **greater** raw force output than electromechanical actuators. However, they carry a **higher** risk of fluid leakage, whereas electromechanical actuators are cleaner and safer to maintain.
5. Relational (SQL) vs. Document (NoSQL) Databases (schema rigidity, horizontal query throughput).
   Ans: Relational databases have a **more rigid** schema than document databases, which allow flexible and evolving data structures. NoSQL databases typically achieve a **higher** horizontal query throughput than relational databases because data can be distributed across many nodes more easily.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair into a single sentence using a defining or non-defining relative clause.

1. (a) The algorithm analyzes microscopic cracks in turbine blades. (b) It was patented by the aeronautics team.
   Ans: The algorithm **that** analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. (a) The lead chemical engineer authorized the emergency boiler shutdown. (b) She noticed an abnormal pressure spike in chamber two.
   Ans: The lead chemical engineer, **who** noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. (a) The laboratory houses a 400 kV impulse voltage generator. (b) High-voltage insulator puncture tests are conducted there.
   Ans: The laboratory, **where** high-voltage insulator puncture tests are conducted, houses a 400 kV impulse voltage generator.
4. (a) We engaged an external cybersecurity consultant. (b) His firmware vulnerability audit exposed three backdoor ports.
   Ans: We engaged an external cybersecurity consultant **whose** firmware vulnerability audit exposed three backdoor ports.
5. (a) The supervisory system utilizes an artificial neural network. (b) It predicts transformer insulation life based on dissolved gas concentrations.
   Ans: The supervisory system utilizes an artificial neural network **that** predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the conditional statements logically and grammatically (model answers).

1. 1(a) If the operating temperature exceeds 85°C, ____
   Ans: …the cooling system **will switch** to high-speed mode and an alarm **will sound**.
2. 1(b) If the operator had pressed the emergency shutdown switch immediately, ____
   Ans: …the machine **would have stopped** before the damage **occurred**.
3. 2(a) If the supply voltage drops below the 200V threshold, ____
   Ans: …the under-voltage relay **will disconnect** the load to protect the motor.
4. 2(b) If the network administrator had enabled multi-factor authentication, ____
   Ans: …the attacker **would not have gained** access to the server.
5. 3(a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, ____
   Ans: …the exhaust fans **will start** automatically and the evacuation alarm **will be triggered**.
6. 3(b) If the mechanical team had checked the bearing lubrication schedule, ____
   Ans: …the bearing **would not have overheated** and failed.
7. 4(a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, ____
   Ans: …it **will stop** immediately and **reroute** along a clear path.
8. 4(b) If the thermal sensor had registered the runaway reaction in time, ____
   Ans: …the control system **would have shut down** the reactor and prevented the explosion.
9. 5(a) If the software receives an unauthenticated API token, ____
   Ans: …it **will reject** the request and **return** an authentication error.
10. 5(b) If the civil inspectors had reviewed the core sample density tests, ____
    Ans: …they **would have identified** the weak concrete before approving the structure.

Exercise 5: Impersonal Passive Voice

Convert the active sentences into the Impersonal Passive Voice for formal engineering reports.

1. “The chemical engineers distilled the crude extract at a constant temperature of 60°C.”
   Ans: The crude extract **was distilled** at a constant temperature of 60°C.
2. “The metallurgist polished the cross-section of the weld seam with fine diamond paste.”
   Ans: The cross-section of the weld seam **was polished** with fine diamond paste.
3. “The quality control division rejected fifteen cast-iron pump casings due to internal porosities.”
   Ans: Fifteen cast-iron pump casings **were rejected** due to internal porosities.
4. “The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter.”
   Ans: The resistance across the bridge circuit **was measured** using a calibrated digital ohmmeter.
5. “The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters.”
   Ans: The emergency aerodynamic braking flaps **were initiated** at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the direct queries into Reported (Indirect) Speech.

1. The safety supervisor asked the intern, “Did you verify the grounding connections on the generator?”
   Ans: The safety supervisor asked the intern **whether he/she had verified the grounding connections on the generator**.
2. The plant director asked the site engineer, “Has the emergency effluent treatment plant received clearance?”
   Ans: The plant director asked the site engineer **whether the emergency effluent treatment plant had received clearance**.
3. The IT security lead enquired of the developer, “Why did you disable the firewall rule on the database server?”
   Ans: The IT security lead enquired of the developer **why he/she had disabled the firewall rule on the database server**.
4. The client asked the project manager, “Can the bridge structure withstand Category 4 hurricane winds?”
   Ans: The client asked the project manager **whether the bridge structure could withstand Category 4 hurricane winds**.
5. The research guide asked the postgraduate scholar, “How many sample trials did you conduct under cryogenic conditions?”
   Ans: The research guide asked the postgraduate scholar **how many sample trials he/she had conducted under cryogenic conditions**.

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the errors involving mass/uncountable nouns.

1. 1(a) We need to install several new software and machineries in the CAD laboratory.
   Ans: Error: *several new software, machineries*. Correct: We need to install **new software and machinery** (or “several new software packages and pieces of machinery”) in the CAD laboratory.
2. 1(b) The team collected plenty of useful informations from the field test.
   Ans: Error: *informations*. Correct: The team collected plenty of useful **information** from the field test.
3. 2(a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.
   Ans: Error: *equipments* (and *apparatuses* as a mass noun). Correct: The technician ordered ten new **pieces of apparatus** and modern electronic **equipment** for the antenna lab.
4. 2(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.
   Ans: Error: *infrastructures, powers*. Correct: High-speed train operations require advanced transport **infrastructure** and dedicated electrical **power**.

5. 3(a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.
   Ans: Error: *datas, pollutions*. Correct: The environmental sensor gathers extensive atmospheric **data** and **pollution** indices every minute.
6. 3(b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.
   Ans: Error: *sands, cements*. Correct: The construction crew unloaded tons of **sand, cement,** and steel debris at the worksite.
7. 4(a) All electrical hardwares and wirings must comply with local safety legislations.
   Ans: Error: *hardwares, wirings, legislations*. Correct: All electrical **hardware and wiring** must comply with local safety **legislation**.
8. 4(b) The data scientist discovered several interesting knowledges after running the machine learning clustering.
   Ans: Error: *knowledges*. Correct: The data scientist discovered several interesting **insights** (or "a great deal of new knowledge") after running the machine learning clustering.
9. 5(a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.
   Ans: Error: *staffs, glasswares, wastes*. Correct: Laboratory **staff** must clear away all **glassware** and acidic **waste** before leaving the premises.
10. 5(b) The project failed due to a lack of proper fundings and technical researches.
    Ans: Error: *fundings, researches*. Correct: The project failed due to a lack of proper **funding** and technical **research**.

Exercise 8: Infinitives of Purpose & Gerund Constructions

Rewrite the pairs using an infinitive of purpose (to + verb) or a gerund after a preposition (by / for + -ing).

1. 1(a) A pressure relief valve is installed. It vents excess boiler steam.
   Ans: A pressure relief valve is installed **to vent** excess boiler steam.
2. 1(b) The automated script runs every midnight. It archives legacy server logs.
   Ans: The automated script runs every midnight **to archive** legacy server logs.
3. 2(a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.
   Ans: Automotive engineers installed an intercooler **to reduce** the compressed air intake temperature.
4. 2(b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.
   Ans: The technicians analyzed the structural stress points **by placing** piezoelectric strain gauges along the girder.
5. 3(a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.
   Ans: High-performance heatsinks feature copper heatpipes **to conduct** heat away from the silicon die.
6. 3(b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.
   Ans: The factory cut water consumption by 30% **by implementing** a closed-loop recycling filtration system.
7. 4(a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.
   Ans: Safety guards are mounted over rotating flywheel belts **to prevent** accidental worker contact.
8. 4(b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.
   Ans: The algorithm optimizes memory bandwidth **by pre-fetching** sequential array indices into L2 cache.
9. 5(a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.
   Ans: The electronics lab installed an uninterruptible power supply (UPS) **to protect** servers against sudden voltage sags.

10. 5(b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.
    Ans: The metallurgy group enhanced surface hardness **by diffusing** nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms.

1. 1(a) The drone adjusted its flight path _____ (automatic) to avoid overhead power cables.
   Ans: automatically
2. 1(b) The software patch was integrated _____ (flawless) into the production server.
   Ans: flawlessly
3. 2(a) The CNC cutting head traced the complex automotive profile _____ (accurate).
   Ans: accurately
4. 2(b) The turbine governor responded _____ (instant) to the unexpected grid frequency drop.
   Ans: instantly
5. 3(a) The hazardous gas extraction vents operate _____ (continuous) throughout the electroplating process.
   Ans: continuously
6. 3(b) Technicians handled the radioactive isotope capsule _____ (cautious) using robotic arm manipulators.
   Ans: cautiously
7. 4(a) The optical pyrometer recorded the furnace temperature _____ (consistent) across three production shifts.
   Ans: consistently
8. 4(b) The structural frame buckled _____ (catastrophic) under the simulated earthquake test load.
   Ans: catastrophically
9. 5(a) The firmware routine checks for incoming serial packets _____ (periodic) every hundred microseconds.
   Ans: periodically
10. 5(b) The titanium alloy workpiece was machined _____ (smooth) to a mirror finish.
    Ans: smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition

Insert an appropriate modal auxiliary conveying the meaning in brackets.

1. 1(a) The microcontroller _____ (capability) drive up to six stepper motors simultaneously.
   Ans: can
2. 1(b) Laboratory staff _____ (strict prohibition) handle corrosive reagents without full face shields.
   Ans: must not (also: may not / cannot)
3. 2(a) Before entering the cleanroom chamber, all personnel _____ (mandatory obligation) put on antistatic footwear.
   Ans: must (also: have to)
4. 2(b) Given the high ambient humidity, water droplet condensation _____ (moderate possibility) form inside the optical housing.
   Ans: may / might
5. 3(a) Unauthorized maintenance personnel _____ (strict prohibition) override the high-voltage interlock switch.
   Ans: must not (also: may not / cannot)

6. 3(b) The upgraded wireless transmitter _____ (past capability) broadcast telemetry signals up to forty kilometers.
   Ans: could
7. 4(a) Quality inspection teams _____ (strong recommendation / advice) recalibrate measuring calipers every ninety days.
   Ans: should
8. 4(b) The digital ammeter is reading zero amperes; the auxiliary fuse _____ (logical deduction / certainty) be blown.
   Ans: must
9. 5(a) Industrial robots _____ (capability) perform repetitive spot welds with sub-millimeter precision.
   Ans: can
10. 5(b) Visitors _____ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.
    Ans: must not (also: may not / cannot)

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

Question 1: Write eight imperative safety instructions for a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

Model Answer

SAFETY INSTRUCTIONS – CHEMISTRY & MATERIALS TESTING LABORATORY

1. Wear a lab coat, splash-proof goggles, nitrile gloves, and closed-toe shoes at all times.
2. Handle volatile, toxic, or concentrated chemicals only inside a working fume hood; never smell or taste any substance.
3. Always add acid to water slowly while stirring; never add water to acid.
4. Label every container clearly and never return unused chemicals to the stock bottles.
5. Do not pipette by mouth; use a pipette filler or a dispenser.
6. Clean all glassware and testing instruments before and after use, and wipe up spills immediately to prevent contamination.
7. Dispose of chemical waste only in the designated, labelled containers; never pour it down the sink.
8. Locate the eyewash station, safety shower, and fire extinguisher before starting work, and report every spill or burn to the instructor at once.

Question 2: Write eight imperative safety instructions for a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.

Model Answer

SAFETY INSTRUCTIONS – ELECTRICAL MACHINES & DRIVES LABORATORY

1. Do not switch on the supply until the instructor has inspected and approved your circuit.
2. Wear insulated footwear and safety glasses, and remove all metallic jewellery and watches before starting.
3. Make or change every connection only when the supply is switched off and isolated.
4. Use only properly rated, insulated leads and check them for damage before use.
5. Ensure that the machine frame is earthed and that all couplings and rotating parts are guarded before loading.
6. Start the motor at the minimum starter/rheostat setting, increase the load gradually, and never exceed the rated current or voltage.
7. Never touch live terminals or rotating parts, keep the floor dry, and never work alone.
8. In an emergency, press the emergency stop, switch off the main supply, and use a CO_2 extinguisher (never water) on electrical fires.

Question 3: Write eight imperative safety guidelines for an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Model Answer

SAFETY GUIDELINES – ADVANCED MANUFACTURING & CNC MACHINING WORKSHOP

1. Wear safety glasses, safety shoes, and close-fitting clothes; tie back long hair and remove rings, watches, and loose gloves.
2. Close the guard door before starting the spindle, and never bypass the safety interlocks.
3. Clamp the workpiece and tool firmly, and remove chuck keys and wrenches before pressing the start button.
4. Keep your hands away from the spindle and cutter head, and wait until all motion stops before measuring or adjusting.
5. Never remove hot chips by hand; use a brush or chip hook after the machine stops, and wear a face shield when chips are flying.
6. Wipe up coolant spills immediately and keep the floor dry and free of oil to prevent slipping.
7. Verify the program and tool offsets with a dry run at low feed rate before cutting.
8. Know the location of the emergency stop button, never leave a running machine unattended, and report any malfunction or injury to the supervisor immediately.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

Question 1: Write a descriptive essay (250 words) on "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit."

Model Answer

Pongal: A Harvest Festival in Harmony with Nature

Pongal, the harvest festival of Tamil Nadu, is celebrated every January to thank the Sun, the earth, and the cattle for a bountiful harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal, and Kaanum Pongal. Villages wake before dawn, houses are cleaned, and colourful kolams drawn with rice flour decorate every doorstep.

The heart of the celebration is the cooking of fresh rice with milk and jaggery in a new clay pot in the open courtyard. When the milk boils over, families shout "Pongalo Pongal!" and offer the dish to the Sun. The festive meal includes sweet pongal, venn pongal, sugarcane, bananas, and sambar served on banana leaves. On Mattu Pongal, cattle are bathed, painted, and garlanded, while neighbours gather for folk songs, kabaddi, and village games that strengthen community bonds.

Modern life, however, has added plastic, smoke, and waste to the celebration. Eco-friendly practices can preserve its core spirit. Clay pots, banana leaves, and areca-leaf plates can replace plastic and thermocol. Kolams made of rice flour feed ants and birds. Instead of burning old tyres and plastics in Bhogi bonfires, communities can collect old items for recycling and compost dry leaves. Community kitchens can use biogas or efficient stoves, and leftover food can be composted or shared with the needy. Decorations of mango leaves, flowers, and sugarcane are fully biodegradable.

By blending tradition with sustainability, Pongal can continue to honour nature while protecting it. A festival that celebrates the earth must, above all, take care of it.

Question 2: Write a descriptive essay (250 words) on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution."

Model Answer

The Temple Car Festival: Faith with Responsibility

The annual temple car (Therottam) festival in my hometown is the most awaited event of the year. Every spring, thousands of devotees gather outside the ancient temple to pull the massive wooden chariot carrying the festival deity through the four mada streets. Days before the event, the temple is whitewashed, the streets are decorated with kolams and banana trees, and the town resonates with nadaswaram music and chanting.

On the main day, the idol is adorned with silk and jasmine garlands and placed on the car. Men, women, and children hold the thick ropes and pull together, chanting in unison, which symbolises unity and collective devotion. Along the route, families offer coconuts, camphor, and fruits, while volunteers distribute prasadam such as pongal, puliyodarai, sundal, and panakam on areca-leaf plates. Small fairs sell toys, bangles, and sweets.

Unfortunately, the festival also generates plastic cups, packets, and loudspeaker noise. Several measures can reduce this impact. The temple trust can ban single-use plastics and provide steel tumblers, banana leaves, and refill-based drinking water points. Volunteer teams can segregate waste at collection points and compost food scraps. Noise can be controlled by limiting loudspeaker decibel levels, replacing firecrackers with traditional instruments, and keeping cultural programmes within fixed hours. Crowd management can be improved with barricaded lanes, trained volunteers, one-way movement, and clearly marked medical and water points.

By combining faith with responsibility, the Therottam can retain its spiritual grandeur while keeping the town clean, peaceful, and safe for future generations.

Question 3: Write a descriptive essay (250 words) on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting."

Model Answer

Folk Art Fairs: Heritage Powered by the Sun

Every winter, our town hosts a regional folk art and handicraft fair where the streets become a living museum of heritage. Stalls display Thanjavur paintings, Kalamkari fabrics, terracotta figures, palm-leaf baskets, and brass lamps. Artisans demonstrate weaving, pottery, and wood carving, while folk groups perform Karagattam, Oyilattam, and Therukoothu in the evenings. The air is filled with drumbeats, the smell of earthen lamps, and the laughter of children, creating a warm and festive atmosphere.

Food is equally central to the experience. Visitors queue for heritage delicacies such as kambu koozh, ragi kali, kuzhi paniyaram, palm-jaggery sweets, murukku, and filter coffee served in steel tumblers. Millets and local produce are promoted in place of industrial snacks.

Modern environmental strategies can make such fairs sustainable. A renewable solar micro-grid, built with rooftop panels and battery storage, can power the stalls, LED lights, and sound systems without noisy diesel generators, cutting emissions and noise. Surplus energy can even be shared with nearby homes. Zero-waste composting is equally important: food stalls can use leaf plates and clay cups, and all leftovers can be collected in colour-coded bins and sent to community compost pits that produce manure for local gardens. Dry waste can be recycled, and water refill stations can replace plastic bottles.

Such measures also educate visitors by showing that tradition and technology can work together. When artisans' skills are supported by clean energy and responsible waste management, folk culture is preserved while the environment is protected.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

Question 1: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam."

Model Answer

Technical Study Tour: Mettur Dam Hydroelectric Station

Route and logistics: Our group of forty second-year students will leave Chennai by a chartered bus at 5:00 a.m. and travel about 340 km via Vellore and Salem to Mettur, reaching the dam by early afternoon. The return journey will be the next evening. Accommodation will be in a nearby guest house, and prior permission from the dam authorities will be obtained for the technical visit.

Technical observations: Built across the Cauvery River in 1934, the Mettur Dam forms the vast Stanley Reservoir. From the dam crest, we will observe the masonry gravity structure, the 16-gate spillway, and the intake towers. Engineers will take us through the powerhouse, where water flowing through penstocks spins the turbines and generators. We will study the head, flow rate, governors, transformers, and the switchyard that transmits power to the grid. The surrounding hills and the calm reservoir offer a perfect view of how water management and power generation coexist.

Local food: After the visit, we will taste the region's famous Mettur fish fry beside the dam, along with Kongunadu-style meals of ragi kali, kambu koozh, and spicy country-chicken gravy. The next morning, we will enjoy Salem's famous mangoes and soft idlis with sambar.

Safety: Students must wear closed shoes, follow the guide's instructions, and avoid restricted zones.

This tour will connect classroom theory on fluid mechanics and power generation with real-world practice.

Question 2: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm."

Model Answer

Technical Study Tour: Wind–Solar Hybrid Energy Farm, Muppandal

Route and logistics: Our team of forty students will board an overnight train from Chennai Egmore to Nagercoil, a journey of about twelve hours. From there, a hired mini-bus will take us roughly 25 km to the wind–solar hybrid farm near Muppandal, one of India's oldest wind-power regions. Accommodation will be arranged at a budget lodge in Nagercoil, and entry permission will be taken in advance.

Technical observations: The Aralvaimozhi gap channels strong winds between the hills, so hundreds of turbines turn in steady rhythm across the landscape. We will study the tower, nacelle, gearbox, blade pitch control, and the generator. Beside them, rows of photovoltaic arrays convert sunlight directly into electricity, and inverters and transformers condition the power before it enters the grid. The key observation will be how wind peaks in the afternoon while solar peaks at noon, so the hybrid system delivers steadier output, supported by an energy management system.

Local food: Along the way, we will sample Nanjil Nadu specialities such as fish curry meals, adai with aviyal, and crisp banana chips, and we will carry Tirunelveli halwa home as a sweet memory.

Safety: Everyone must wear helmets and safety shoes, stay within marked areas, and avoid going near rotating blades or electrical yards.

This tour will convert our knowledge of renewable energy into practical understanding.

Question 3: Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant."

Model Answer

Field Visit: Seawater Reverse Osmosis Desalination Plant, Nemmeli

Route and logistics: Our group of forty students will leave the college campus at 8:00 a.m. in two hired buses and travel along the scenic East Coast Road to the Nemmeli desalination plant, about 35 km away. The visit is scheduled from 10:30 a.m. to 1:30 p.m., after prior written approval from the plant authorities. We will return by 5:00 p.m.

Technical observations: At the *intake pumping station*, seawater is drawn through submerged pipes and screens. We will follow it through pre-treatment filters that remove sand and suspended particles, and then to the long *membrane filtration racks*, where high-pressure pumps force water through semi-permeable membranes, leaving the salts behind. Energy recovery devices recapture pressure from the reject stream and cut power consumption. Finally, the *brine discharge outlets* release the concentrated salt stream through diffusers into the sea, where it is diluted to protect marine life. A SCADA control room monitors the pressure, flow, and conductivity automatically.

Coastal food experience: On the way back, we will stop at a seaside restaurant for fried fish, prawn masala, and crab curry with steamed rice, followed by tender coconut and hot filter coffee.

Safety: Students must wear closed shoes and helmets, keep away from high-pressure lines, and follow the guide at all times.

This visit will show how engineering can turn seawater into a reliable source of drinking water.

Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

Question 1: Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon.

Model Answer

[Your Name]
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to use the computer laboratory after college hours for National AI Hackathon preparation

Respected Sir/Madam,

We, a team of four second-year students, have been shortlisted for the National AI Hackathon to be held on 20 and 21 December 2026. We are developing an AI-based system that detects crop diseases from leaf images using a convolutional neural network, with the aim of helping farmers receive quick and accurate advice.

To prepare effectively, our team needs to train and test the model on a high-performance system, which is not available in our hostels. We therefore request permission to use the Computer Science laboratory from 5:00 p.m. to 8:00 p.m. on weekdays from 10 November to 15 December 2026. We will require: (1) GPU-enabled workstations, (2) Python, TensorFlow, PyTorch, and Jupyter Notebook, (3) high-speed internet access to download datasets, and (4) access to the project server for version control.

We assure you that we will follow all laboratory rules, handle the equipment carefully, and switch off all systems before leaving. If required, a faculty member may supervise our sessions.

We kindly request you to grant us permission. We shall be grateful for your support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of Team [Team Name])
Register No.: [____]

Question 2 (Industrial Visit): Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.

Model Answer

The Head, Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Plant Manager
[Name of Factory], Industrial Robotics & Automation Division
[Address of Factory]

Subject: Request for permission to conduct an industrial visit for first-year engineering students

Respected Sir/Madam,

I am writing on behalf of the Department of Mechanical Engineering, St. Joseph's College of Engineering, to request your kind permission for 50 first-year engineering students, accompanied by two faculty members, to visit your robotics and automation plant on any convenient weekday between 15 and 30 November 2026, from 10:00 a.m. to 1:00 p.m.

The visit aims to give our students exposure to modern industrial practice. The intended learning outcomes are: (1) to understand how robotic arms, PLCs, and sensors are used in automated assembly lines; (2) to observe quality-control and safety systems in a real production environment; (3) to relate classroom concepts of mechanics, electronics, and programming to industry; and (4) to learn about career opportunities in automation and robotics.

We will strictly follow your safety rules and dress code, and the students will remain under faculty supervision at all times. We would be grateful if you could also depute an engineer to brief the students.

We look forward to your favourable reply and thank you for your time and support.

Yours faithfully,
[Name]
Head of the Department
Contact: [phone / email]

Question 3 (Practice A – Workshop Facilities): Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge.

Model Answer

[Your Name]
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for machine shop access after college hours for fabrication of an electric go-kart

Respected Sir/Madam,

Our six-member student team has registered for the Inter-Collegiate Design Challenge to be held in January 2027. We plan to design, fabricate, and test an energy-efficient electric go-kart powered by a brushless DC motor and a lithium-ion battery pack, with a lightweight steel-tube chassis to maximize range.

Since regular class hours do not allow enough time for fabrication, we request permission to use the machine shop from 4:30 p.m. to 7:30 p.m. on weekdays and on Saturdays from 9:00 a.m. to 1:00 p.m., from 10 November 2026 to 10 January 2027. Our requirements are: (1) welding and cutting equipment, (2) lathe and drilling machines, (3) a workbench and measuring tools, and (4) space to assemble and test the vehicle.

We promise to wear proper safety gear, to operate the machines only under the technical staff's supervision, and to keep the workshop clean. We will also share our design reports with the department.

We shall be highly obliged if you would kindly grant us permission.

Thanking you,

Yours faithfully,
[Your Name]
Team Captain, Team [Name]
Register No.: [____]

Question 4 (Practice B – Sponsorship & Duty Leave): Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Model Answer

[Your Name]
Department of [____]
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

We are happy to inform you that our five-member multidisciplinary team (Mechanical, Electronics, Computer Science, and Aeronautical students) has been selected to present its AI-assisted drone surveillance prototype at the National Defense Innovation Expo, to be held in New Delhi from 12 to 14 December 2026.

The prototype uses a quadcopter with a thermal camera and an onboard object-detection algorithm to monitor border-like terrain in real time. Presenting it at the Expo will give our college national visibility and help us receive expert feedback.

We therefore request (1) **duty leave** for all five members from 10 to 16 December 2026, including travel days, and (2) **financial sponsorship** of approximately Rs. 60,000, covering train fares, accommodation, registration fees, and transportation of the prototype. A detailed budget is attached.

We assure you that we will make up for missed classes and will submit a report after the Expo.

We shall be extremely grateful for your kind consideration and support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of the team)
Enclosure: Selection letter and budget estimate

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions to evaluate technical trade-offs.

Question 1: Compare and contrast the positives and negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles.

Model Answer

Electric Vehicles versus Internal Combustion Engine Vehicles

Electric vehicles and conventional ICE vehicles represent two very different approaches to personal transport, each with clear strengths and weaknesses.

In terms of battery life and fuel, an EV battery typically lasts eight to ten years before its capacity falls noticeably, and replacement is costly. An ICE vehicle, on the other hand, relies on an engine that can last far longer with regular servicing, although it needs constant fuel. Regarding the manufacturing footprint, EVs have a higher production impact because mining lithium, cobalt, and nickel for batteries is energy-intensive. In contrast, ICE vehicles have a lower manufacturing footprint but produce continuous tailpipe emissions throughout their life, whereas EVs emit none while driving.

Operating costs favour EVs. Electricity is cheaper per kilometre than petrol or diesel, and EVs have fewer moving parts, so servicing costs less. Conversely, ICE vehicles need oil changes, filters, and more frequent repairs, and they are exposed to fluctuating fuel prices. However, the purchase price of an EV is still higher than that of a comparable ICE vehicle.

Charging network feasibility remains the biggest challenge for EVs. Although charging stations are growing in cities, they are scarce on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, by contrast, enjoy a mature and widespread fuel network and refill within minutes.

In conclusion, EVs are better for the environment and cheaper to run, whereas ICE vehicles are more convenient and cheaper to buy today. As charging networks expand and battery costs fall, the balance is likely to shift towards electric mobility.

Question 2: Compose an analytical essay comparing the positives and negatives of Remote/Virtual Internships versus On-Site Physical Internships for early-stage engineering students.

Model Answer

Remote Internships versus On-Site Internships

For early-stage engineering students, internships provide the first glimpse of professional life. Remote and on-site internships both offer value, but in different ways.

Regarding hands-on skill acquisition, on-site internships are clearly superior. Students can handle real machines, instruments, and prototypes, and they learn workshop discipline. Remote internships, in contrast, depend mainly on simulations, software, and coding tasks, so they suit computing and design roles but limit practical exposure in mechanical, electrical, or civil fields.

In terms of schedule flexibility, remote internships have a distinct advantage. Students can balance them with classes, avoid travel and accommodation costs, and even work for companies in other cities. On-site internships, on the other hand, require fixed hours, relocation, and higher expenses, which may disrupt academic schedules.

When it comes to mentorship, on-site interns benefit from direct supervision, spontaneous questions, and immediate feedback. Remote mentoring, whereas, often relies on scheduled video calls, which can leave students feeling isolated or reluctant to ask doubts.

Professional networking follows a similar pattern. Physical presence helps students build relationships with colleagues, observe workplace culture, and receive recommendations. Virtual interns conversely meet fewer people, and their connections tend to remain limited.

In conclusion, on-site internships are more effective for practical skills, mentorship, and networking, whereas remote internships are more flexible and economical. A hybrid model that combines both may offer first-year engineers the best balance of learning and convenience.

Question 3: Write an analytical Compare and Contrast essay (250–300 words) evaluating the positives and negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises.

Model Answer

Cloud Computing versus On-Premise Data Centers

Engineering enterprises must decide whether to host their applications and data on the cloud or in their own physical data centres. Each option carries distinct advantages and drawbacks.

With respect to upfront capital expenditure, the cloud is clearly attractive. Companies pay only for the resources they use, so they avoid buying servers, cooling systems, and real estate. On-premise data centres, in contrast, demand a large initial investment before a single application runs.

Recurring overheads, however, tell a different story. Cloud subscription fees can accumulate quickly for heavy, continuous workloads, and unexpected usage can inflate bills. On the other hand, once an on-premise facility is built, its running costs are more predictable, although the enterprise still pays for electricity, maintenance, and skilled staff.

Data security and compliance are two sides of the same coin. Leading cloud providers invest heavily in encryption and certified security, but sensitive design files or defence-related data may be subject to rules requiring local storage. On-premise systems give the enterprise full control over its data; conversely, they place the entire burden of protection on the company's own staff.

Workload scalability is the cloud's greatest strength. Computing power for simulations or testing can be increased within minutes and reduced afterwards. Physical data centres, whereas, are limited by installed hardware, and expanding them takes weeks or months.

In summary, the cloud offers flexibility and low initial cost, whereas on-premise data centres offer control and predictable long-term expenses. Many enterprises now adopt a hybrid approach to enjoy the benefits of both.

Question 4: Write a balanced Compare and Contrast essay (250–300 words) discussing the positives and negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication.

Model Answer

3D Printing versus CNC Milling in Prototype Fabrication

Additive manufacturing (3D printing) builds parts layer by layer, whereas subtractive CNC milling carves them from a solid block. Both are widely used for prototyping, and each has its own trade-offs.

In terms of material wastage, 3D printing has a distinct advantage because it uses only the material needed for the part and its supports. CNC milling, in contrast, removes up to 90 percent of the block as chips, although this metal can often be recycled.

Regarding production speed for batch runs, the picture reverses. A 3D printer may take many hours to complete a single part, so printing large batches is slow. On the other hand, once a CNC machine is programmed, it can produce many identical parts quickly and consistently, making it more efficient for batch production.

Surface finish and accuracy favour CNC milling. It achieves tight tolerances and smooth surfaces directly from the machine, and it works with strong engineering metals. 3D-printed parts, conversely, often show visible layer lines and may need sanding or post-processing to meet precision requirements.

Geometric complexity is where 3D printing shines. It can create internal channels, lattice structures, and organic shapes that are impossible or very expensive to machine. CNC milling, however, is limited by tool access, so deep cavities and undercuts require special setups or multiple operations.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with minimal waste, whereas CNC milling is better for accurate, smooth, and repeatable parts. Choosing between them depends on the design complexity, quantity, and finish required.

Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

Question 1: Draft a personal email to your high school mentor or teacher sharing your first-semester experiences at engineering college.

Model Answer

To: mentor.name@example.com
Subject: My First Semester at Engineering College – Thank You for Your Guidance

Dear Sir/Madam,

I hope you are doing well. I am writing to share my first-semester experiences at St. Joseph's College of Engineering and to thank you for the guidance that helped me reach here.

The biggest change has been the transition to independent campus life. Living in the hostel has taught me to manage my own time, laundry, and budget, and I have made good friends from different states. At first, I missed home, but a regular routine and weekly calls to my parents have helped.

Academically, I have enjoyed the laboratory sessions the most. In the Basic Electrical lab, I built my first series–parallel circuit, and in the programming lab, I wrote a small C program to calculate the average marks of a class. Working in teams has taught me patience and the value of recording observations carefully.

My end-semester examinations begin next month, so I have prepared a study plan. I revise two subjects a day, solve previous question papers on weekends, and practise numerical problems every evening. I have also formed a small study group to clear doubts.

Your advice to "stay curious and consistent" has stayed with me, and I would be grateful for any further suggestions you have for me. I would love to visit the school during the holidays.

With warm regards,
[Your Name]

Question 2: Draft a personal email to your high school science teacher reflecting on your first term in engineering, focusing on hands-on lab coursework, your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.

Model Answer

To: science.teacher@example.com
Subject: Reflections on My First Term in Engineering

Dear Ma'am/Sir,

I hope you and everyone at school are well. It has been almost four months since I joined engineering, and I wanted to share how things have been going, since your science classes inspired my choice.

Adapting to hands-on lab coursework was exciting but challenging. Unlike school experiments, we now work with oscilloscopes, breadboards, and microcontrollers, and a small wiring error can ruin the whole circuit. I have learned to read the manual first, take careful readings, and write neat observations, which you always insisted on.

I also joined the college's Robotics Club and, last month, took part in my first hackathon. Our team built a simple line-following robot in 24 hours. We did not win, but we learned about teamwork, debugging, and handling pressure.

Now I am preparing for the university theory and practical examinations. My strategy is to divide each syllabus into units, revise one unit every day, and solve past question papers. For practicals, I am repeating every experiment and revising the viva questions with my lab partner.

I would be glad to hear any advice you may have. Please convey my regards to all the teachers.

Yours sincerely,
[Your Name]

Question 3: Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs, describing your daily schedule, challenges in managing lab record submissions, and asking for tips on semester exam revision.

Model Answer

To: elder.sibling@example.com
Subject: Need Your Advice on Balancing Studies and Clubs

Hi Anna/Akka,

How are you? I hope work and life are going well. I am writing because I badly need your advice, since you handled college life so well.

As you know, I recently joined the IEEE student chapter and the coding club. My day starts at 6:30 a.m.; classes run from 9:00 to 4:30, and the club meets from 5:00 to 6:30 p.m. After dinner, I study for two hours or work on projects. I enjoy everything, but I often feel tired and short of time.

My biggest challenge is managing lab record submissions. Each experiment needs a neat record with circuit diagrams, observations, and results, and I usually postpone writing until the deadline, which creates last-minute stress. Sometimes I also forget the viva questions.

Could you please tell me how you balanced clubs and studies? Should I reduce my club hours during exam time? Do you have any tips for writing records regularly, perhaps by finishing them the same day?

Also, my end-semester exams are in a month, and I would love your tips for revision, such as how to make short notes, how many previous papers to solve, and how to handle difficult subjects like Engineering Mathematics.

Thanks a lot in advance. Please give my regards to everyone there, and let's talk this weekend.

Love,
[Your Name]

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

Question 1: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station.

Model Answer

4-Day Technical Tour: Satish Dhawan Space Centre (Sriharikota)

Participants and transport: 40 students and 2 faculty members travel in a chartered AC bus from Chennai (about 100 km), after obtaining official visitor permission and ID verification.

Day 1: Depart at 6:00 a.m., reach Sullurpeta by 9:00 a.m., and check in at a lodge. Afternoon briefing on rocket propulsion and launch operations.
Day 2: Visit the Space Centre: launch pads, the Vehicle Assembly Building, and the launch viewing gallery; interact with scientists about satellite launch vehicles and telemetry.
Day 3: Return to Chennai and visit Birla Planetarium and a science centre to study orbital mechanics and satellite applications.
Day 4: Group discussion, project presentations, and preparation of the technical report.

Budget (per student):
Bus hire (share): Rs. 1,200
Accommodation (2 nights): Rs. 1,600
Food: Rs. 1,200
Entry and permits: Rs. 300
Insurance and contingency: Rs. 300
Total: Rs. 4,600 (about Rs. 1,84,000 for 40 students)

Safety protocols: Carry photo ID; follow security instructions strictly; do not photograph restricted areas; wear closed shoes and stay with the group; maintain a first-aid kit and emergency contact list; and obey the guide at all times.

Question 2: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility.

Model Answer

4-Day Industrial Exposure Tour: Cochin Port and Shipyard, Kochi

Transport arrangements: 40 students and 2 faculty members travel by reserved sleeper-class train from Chennai to Ernakulam (about 12 hours) and use a hired mini-bus for local movement.

Day 1: Overnight journey; arrive next morning, check in at a hostel, and rest.
Day 2: Technical briefing at the port: container terminal, cranes, berthing, and cargo handling systems.
Day 3: Visit to the shipyard's dry dock: ship repair, hull inspection, welding, and propulsion systems; briefing on naval architecture and dock operations.
Day 4: Interaction with port engineers, a short harbour cruise, and an evening return journey.

Budget (per student):
Train fare (return): Rs. 1,600
Stay (3 nights): Rs. 2,400
Food: Rs. 2,000
Local transport: Rs. 800
Entry fees and safety kits: Rs. 400
Contingency: Rs. 300
Total: Rs. 7,500 (about Rs. 3,00,000 for 40 students)

Dockyard safety protocols: Wear helmets, safety shoes, and reflective jackets; stay inside marked walkways; do not go near cranes, edges of the dock, or welding areas; never use phones near hazardous zones; carry life jackets on the harbour cruise; and follow the guide's instructions.

Question 3: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center.

Model Answer

4-Day Technical Study Tour: Chittaranjan Locomotive Works and Railway Signalling Centre

Rail transit bookings: 40 students and 2 faculty members assemble at Howrah, Kolkata, and travel by pre-booked reserved second-class seats on the Chittaranjan line (about four hours), with group tickets arranged through IRCTC 60 days in advance. (The starting point may be altered to suit your location.)

Day 1: Rail journey to Chittaranjan; check in at a guest house; orientation on electric traction.
Day 2: Shop-floor visit: bogie assembly, traction motor winding, transformer shop, and final testing of electric locomotives.
Day 3: Visit to an automated signalling centre: electronic interlocking, centralised traffic control, track circuits, and the control panel room.
Day 4: Technical debrief, quiz, and return journey.

Category-wise expenditure (per student):
Rail fare (return): Rs. 1,000
Accommodation (3 nights): Rs. 2,400
Food: Rs. 1,800
Local transport: Rs. 600
Entry and safety gear: Rs. 400
Contingency: Rs. 300
Total: Rs. 6,500 (about Rs. 2,60,000 for 40 students)

Visitor safety rules: Wear helmets, safety shoes, and ear protection on the shop floor; stay behind yellow lines; never touch overhead lines, traction equipment, or tracks; do not cross rails; avoid photography in restricted areas; and move only with the guide.

St. Joseph's College of Engineering

OMR, Chennai - 119 (An Autonomous Institution)

Professional English - I (Lab Integrated) | EN25C09

Model I Practice Worksheet - Questions and Model Answers

PART A - Short Grammar Questions

15 questions | 1 mark each

1. Fill in the blank with the correct form of the verb: The automated system ___ (monitor) the temperature continuously to ensure safe operation.

Ans: monitors.

2. Fill in the blank with the correct past form: The engineers ___ (record) the vibration readings during yesterday's trial.

Ans: recorded.

3. Use the comparative form: The newly developed composite material is significantly ___ (strong) than the conventional material while maintaining a lower weight.

Ans: stronger.

4. Choose must/can/might: Engineers ___ follow safety regulations while operating heavy machinery.

Ans: must.

5. Choose a modal expressing capability: Modern sensors ___ detect even minor changes in temperature.

Ans: can.

6. Combine using a suitable relative pronoun: The laboratory developed a new testing device. The device measures material strength with high precision.

Ans: The laboratory developed a new testing device that measures material strength with high precision.

7. First conditional: If the temperature ___ (rise) above the permissible limit, the cooling system will activate automatically.

Ans: rises.

8. Third conditional: If the technician had inspected the circuit earlier, the equipment ___ (not / fail) during operation.

Ans: would not have failed.

9. Combine using an infinitive of purpose: The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Ans: The mechanics installed dual damper springs to isolate chassis resonance.

10. Use the gerund after the preposition: The research team reduced energy consumption by ___ (optimize) the motor control algorithm.

Ans: optimizing.

11. Convert to impersonal passive: The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Ans: Fifty millilitres of hydrochloric acid was poured into the titration flask.

12. Use the adverb from smooth: The robotic arm moves ___ to ensure accurate positioning.

Ans: smoothly.

13. Rewrite in indirect speech: The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Ans: The project supervisor asked the engineer whether he or she had calibrated the pressure sensor before testing.

14. Correct the noun form: The laboratory purchased several new equipments for conducting material testing.

Ans: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Use the correct irregular comparative: Because the heat sink had poor thermal paste contact, heat dissipation was ___ (bad) than expected.

Ans: worse.

PART B - Applied Grammar and Writing

10 questions | 2 marks each

1. Rewrite in the simple past: The engineer calibrates the pressure sensor and verifies its accuracy.

Ans: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and desktop in portability and screen size.

Ans: Portability: A laptop is more portable than a desktop computer.
Screen size: A desktop computer usually has a larger screen than a laptop.

3. Combine with a relative clause: The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Ans: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Complete logically: (a) If the battery voltage drops below the required level, ... (b) If the technician had checked the connections earlier, ...

Ans: (a) If the battery voltage drops below the required level, the warning indicator will light up.
(b) If the technician had checked the connections earlier, the equipment might not have stopped.

5. Convert to impersonal passive: The laboratory assistant heated the solution to 80 degrees C before testing.

Ans: The solution was heated to 80 degrees C before testing.

6. Rewrite in indirect speech: The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Ans: The project manager asked the technician whether he or she had inspected the pressure valves before starting the compressor.

7. Correct the mass noun errors: (a) The engineer received many feedbacks on the new design. (b) The students received several useful advices from their teacher.

Ans: (a) The engineer received a lot of feedback on the new design. (b) The students received several useful pieces of advice from their teacher.

8. Rewrite using an infinitive of purpose or gerund: (a) The technician wears safety gloves. He wants to protect his hands. (b) The cooling fan operates continuously. It prevents the motor from overheating.

Ans: (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.

9. Use the correct adverb: (a) The machine operates ___ (efficient). (b) The technician checked the electrical connections ___ (careful).

Ans: (a) efficiently (b) carefully.

10. Insert modal auxiliaries for capability and strict prohibition: (a) The robotic arm ___ lift heavy components. (b) Visitors ___ enter the testing area without authorization.

Ans: (a) can (b) must not.

PART C - Extended Writing

The worksheet asks you to answer Question 26 and choose one alternative in Questions 27-30. Both alternatives are provided here for practice.

Question 26 - Internet Safety Instructions

Prompt: Write eight imperative safety instructions for safe and responsible internet use.

Model Answer

1. Use a strong, unique password for every important account.
2. Turn on two-factor authentication whenever it is available.
3. Do not click suspicious links or open unexpected attachments.
4. Verify a website address before entering personal or payment information.
5. Keep your operating system, browser and security software updated.
6. Never share passwords, one-time codes or sensitive personal details with strangers.
7. Use secure networks and avoid sensitive transactions on public Wi-Fi.
8. Report suspicious messages, cyberbullying and suspected account activity to a trusted adult or the relevant service.

Question 27.a - Traditional Art in Tamil Nadu

Prompt: Write about a regional traditional art form, its origins and practices, and ways to preserve it.

Model Answer

Therukoothu, or street theatre, is a traditional performance art of Tamil Nadu. It grew in village communities as a way to share stories, celebrate festivals and bring people together. Many performances draw on episodes from the Mahabharata and other well-known stories. The open-air setting made the art accessible to everyone, without the need for a formal theatre.

A traditional performance combines acting, music, dance, colourful costumes and expressive makeup. Performers sing and speak in Tamil, while drums and other instruments support the rhythm. The actors often use energetic gestures and direct interaction with the audience. Performances may continue for several hours, especially during temple festivals. Artists learn through practice, observation and guidance from experienced performers.

Today, migration, changing entertainment habits and limited income can make it difficult for younger people to continue this demanding art. Schools and colleges can invite performers for demonstrations and workshops. Local cultural groups can record performances, interview senior artists and create subtitled videos for online audiences. Festival organisers can provide fair fees, safe stages and regular performance opportunities. Grants and scholarships can support young learners.

Modern technology can help preserve Therukoothu when it documents and promotes the art without replacing live performance. Young people can learn its music, stories and stage skills while also sharing it through responsible digital media. Supporting the artists and giving them opportunities to teach will keep this cultural tradition meaningful for future generations.

Question 28.a - Request to Extend Library Hours

Prompt: Write a formal letter to the college librarian requesting longer hours during semester examinations.

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Librarian
[College Name]

Subject: Request to extend library working hours during semester examinations

Respected Sir/Madam,

I am writing on behalf of students in our department to request an extension of the college library's working hours during the forthcoming semester examination period. Many students need a quiet place to revise, consult reference books and work on assignments after regular classes. Longer access would particularly help students who cannot study effectively in crowded or noisy accommodation.

We expect many students to use the library, especially in the evenings. We propose extending the closing time by two hours on weekdays during the examination period, subject to staff availability and college approval. Students can sign in with their college identification cards, maintain silence, return books on time and follow all library rules.

The library may keep the reference and study areas open, with a staff member or security officer present. A sign-up sheet can help estimate attendance, and students can volunteer to leave their study spaces tidy. We understand that staffing and safety must be considered, and we would be grateful for any schedule the college can support.

This arrangement would encourage focused study and equal access to academic resources. Thank you for considering our request.

Yours faithfully,
A. Student
Class Representative

Question 29.a - AI-Based and Teacher-Based Assessment

Prompt: Write a 250-300 word analytical comparison addressing accuracy, feedback time, fairness and human judgement.

Model Answer

Assessment helps engineering students understand what they know and what they need to practise. Artificial Intelligence-based assessment and traditional teacher-based assessment can both support learning, but each has strengths and limitations.

AI systems can mark objective questions quickly and apply the same scoring rules to every submission. They can analyse large amounts of work, identify patterns in errors and provide feedback soon after a student completes a task. This speed is useful in large classes, where a teacher may need several days to return work. Automated tools can also offer extra practice tailored to a student's weaknesses.

However, accuracy depends on the quality of the questions, answer keys and training data. An AI system may misread an unusual but valid explanation, reward formulaic responses or perform poorly when students use unfamiliar language. If its design reflects biased data, its results may be unfair to some learners. Students also need clear information about how their work is collected and assessed.

Teachers can interpret context, recognise original reasoning and ask follow-up questions. They can judge laboratory teamwork, design choices and communication in ways that are difficult to reduce to a score. Their feedback may take longer, and personal expectations can sometimes create inconsistency, but professional judgement allows them to explain a decision and respond to a student's needs.

A balanced approach is most effective. AI can assist with routine marking, provide practice and flag work for review. Teachers should set assessment goals, check automated results and make final decisions on complex work. In engineering education, accuracy and fairness require both consistent tools and human oversight. Used transparently, AI can save time while teachers preserve the judgement, encouragement and accountability that good assessment needs.

Question 30.a - Email About First Semester

Prompt: Write a personal email to a school friend about your first semester in engineering.

Model Answer

To: friend@example.com
Subject: My first semester in engineering

Hi Ravi,

How are you? I have been meaning to tell you about my first semester at engineering college. It has been busy, exciting and very different from school.

I have made friends with classmates from several towns. We help one another with notes and work together on small lab tasks. The classroom activities include lectures, demonstrations, group discussions and practical sessions. I especially enjoy seeing how ideas from physics and mathematics are used in experiments.

Adjusting to college life took a few weeks. I had to plan my day, keep track of deadlines and become more independent. The campus is much larger than our school, and I am still discovering useful places to study and relax.

Our semester examinations are coming soon, so I have started revising each subject and solving sample problems. I feel a little nervous, but my study group keeps me motivated. I hope the exams go well and that the next semester brings new projects and experiences.

Write back and tell me how you are doing. I hope we can meet during the holidays.

Best wishes,
Arun